



## 1. RESEARCH QUESTION

- Q1** Does current cigarette use differ between male and female students?
- Q2** Are female smokers behaviorally different from female non-smokers?
- Q3** Do behavioral risk characteristics become more prevalent as smoking intensity increases?

## 2. DATA

Dataset: YRBS 2007

## Sample Size

|                 |        |
|-----------------|--------|
| Total Students  | 14,041 |
| Cleaned Sample  | 13,312 |
| Female Students | 7,036  |

## Key Variables



• CurrentCigaretteUse

## Behavioral Factors

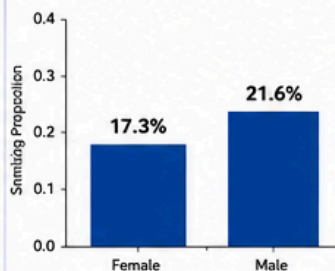
- Alcohol Use
- Binge Drinking
- Marijuana Use
- Physical Fighting
- Sleep
- SadOrHopeless

## 3. EDA HIGHLIGHTS

Gender Difference  
(Notebook 1)

Male students reported higher smoking prevalence.

Current Cigarette Use Proportion by Gender

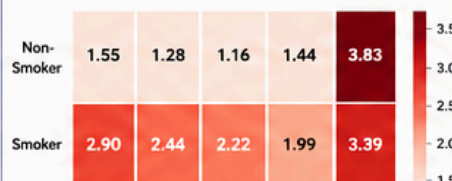


Difference = 4.26%  
(95% CI: 0.029 – 0.056)  
 $z = 6.215, p < 0.001$

Female Smoker Profile  
(Notebook 2)

Compared with female non-smokers, smokers showed higher risk in most behaviors and shorter sleep duration.

## Behavioral Profile of Female Smokers and Non-Smokers

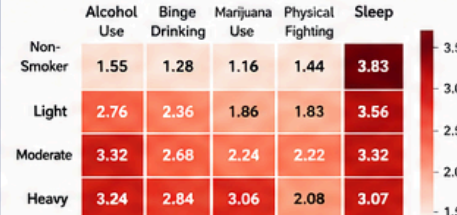


All Mann-Whitney U tests  $p < 0.001$   
SadOrHopeless:  $\chi^2 = 214.61, p < 0.001$

Smoking Intensity Trend  
(Notebook 3)

Risk behaviors generally increased as smoking intensity increased.

## Behavioral Risk Trends Across Smoking Intensity Groups



All Chi-square tests  $p < 0.001$   
Strongest association: Marijuana Use  
( $\chi^2 = 1981.21$ )

## 4. METHODS

## Notebook 1

## Two-Proportion Z-Test

- Compare smoking proportion between Male and Female

## Notebook 2

## Mann-Whitney U Test

- Compare female smokers vs. female non-smokers

## Notebook 3

## Chi-Square Test of Independence

- Compare smoking intensity groups (Non-Smoker, Light, Moderate, Heavy)

## 5. KEY RESULTS

## A. Gender Difference (Notebook 1)

|                            |                |
|----------------------------|----------------|
| Female Proportion          | 17.3%          |
| Male Proportion            | 21.6%          |
| Difference (Male – Female) | 4.26%          |
| 95% Confidence Interval    | (0.029, 0.056) |
| z Statistic                | 6.215          |
| p-value                    | < 0.001        |

Conclusion: Reject  $H_0$ . Male students have a higher smoking proportion.

## B. Female Smoker Profile (Notebook 2)

All behavioral variables (Mann-Whitney U tests)

|                          |             |                     |
|--------------------------|-------------|---------------------|
| Alcohol Use              | $p < 0.001$ | ↑ Alcohol Use       |
| Binge Drinking           | $p < 0.001$ | ↑ Binge Drinking    |
| Marijuana Use            | $p < 0.001$ | ↑ Marijuana Use     |
| Physical Fighting        | $p < 0.001$ | ↑ Physical Fighting |
| Sleep (lower is shorter) | $p < 0.001$ | ↓ SadOrHopeless     |
|                          |             | ↓ Sleep Duration    |



SadOrHopeless (Yes %)  
 $\chi^2 = 214.61, p < 0.001$

## C. Smoking Intensity Trend (Notebook 3)

All Chi-square tests  $p < 0.001$ 

|                   |                    |
|-------------------|--------------------|
| Alcohol Use       | $\chi^2 = 1720.87$ |
| Binge Drinking    | $\chi^2 = 1580.17$ |
| Marijuana Use     | $\chi^2 = 1981.21$ |
| Physical Fighting | $\chi^2 = 298.78$  |
| SadOrHopeless     | $\chi^2 = 190.99$  |

## SadOrHopeless (% Yes)

| Smoking Intensity | % Yes  |
|-------------------|--------|
| Non-Smoker        | 32.82% |
| Light             | 53.15% |
| Moderate          | 56.72% |
| Heavy             | 54.25% |



Risk behaviors generally increase with smoking intensity.  
Strongest upward trend: Marijuana Use. Sleep shows the opposite trend.

## 6. CONCLUSION (KEY TAKEAWAY)

## Gender Difference

Male students showed a significantly higher prevalence of current cigarette use than female students.

## Female Smoker Profile

Female smokers exhibited a broader behavioral and psychological risk profile and reported shorter sleep duration.

## Smoking Intensity Trend

Behavioral risk characteristics generally became more prevalent as smoking intensity increased, while sleep duration decreased.

## Overall Conclusion

Smoking among female students is part of a broader behavioral risk profile. These findings support targeted interventions addressing multiple co-occurring risk behaviors in adolescents.